(CE7456 Lecture Notes 9) Subgraph Enumeration

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Today, we discuss the *subgraph enumeration* problem (also known a *subgraph listing*). Let $G = (V, E)$ be an undirected graph called the *data graph*. A graph $G' = (V', E')$ is a *subgraph* of G if $V' \subseteq V$ and $E' \subseteq E$. Let $Q = (V_Q, E_Q)$ be another undirected graph called the *pattern graph*. A subgraph $G' = (V', E')$ of G is *isomorphic* to Q if there is a bijection function $f : V_Q \rightarrow V'$ such that, for any distinct vertices $u, v \in V_Q$, the pattern graph Q has an edge $\{u, v\}$ if and only if the edge $\{f(u), f(v)\}$ exists in G'. The goal of subgraph enumeration is to report all the subgraphs of G that are isomorphic to Q. Those subgraphs are called the *occurrences* of Q.

We assume that Q has a constant size, i.e., the number $|V_Q|$ can be regarded as a constant (and so can the number $|E_Q|$). Without this assumption, the problem is NP-hard because deciding whether G has a clique is NP-hard.

Math Conventions. For an integer $x \geq 1$, the notation $[x]$ represents the set $\{1, 2, ..., x\}$. The notation $\mathbb{R}_{\geq 0}$ represents the set of non-negative real numbers.

1 Fractional Edge Cover Number

A function $w : E_Q \rightarrow \mathbb{R}_{\geq 0}$ is said to be a *fractional edge cover* of $Q = (V_Q, E_Q)$ if, for every vertex $u \in V_Q$, it holds that

$$\sum_{e \in E_Q : u \in e} w(e) \quad \geq \quad 1. \tag{1}$$

We refer to $w(e)$ as the *weight* of e under w. The *total weight* of w is defined as

$$\sum_{e \in E_Q} w(e).$$

The *fractional edge cover number* of Q — denoted as $\rho(Q)$ — is the smallest total weight of all fractional edge covers of Q. A fractional edge cover is *optimal* if its total weight is $\rho(Q)$.

For example, if Q is a k-cycle (i.e., a cycle with k edges), then $\rho(Q) = k/2$; if Q is a k-star (i.e., a graph with $k + 1$ vertices $u, v_1, ..., v_k$, and k edges $\{u, v_1\}, \{u, v_2\}, ..., \{u, v_k\}$), then $\rho(Q) = k$.

The following is known as the *AGM bound*:

Theorem 1. *If the data graph G has m edges and the pattern graph Q has a constant size, then G can have $O(m^{\rho(Q)})$ occurrences of Q.*

The AGM bound is tight:

Theorem 2. *Fix any integer $m \geq 1$ and any pattern graph Q with a constant size. There is a data graph G with at most m edges that has $\Omega(m^{\rho(Q)})$ occurrences of Q.*

We will prove Theorem 1 in this lecture and Theorem 2 in a future lecture of this course.

2 Enumerating Odd-Length Cycles

Let Q be a k-cycle with an odd number $k = O(1)$. In this section, we give an algorithm to find all its occurrences in a data graph G of m edges in $O(m^{\rho(Q)})$ time where $\rho(Q) = k/2$.

Call a vertex u of G *heavy* if the degree of u is at least $\sqrt{m}$, or *light* otherwise. The number of heavy vertices of G is at most $2\sqrt{m}$ (otherwise, G would have more than m edges).

Let us denote the vertices in Q as $q_1, q_2, ..., q_k$, respectively. To find an occurrence of Q, we need to map each q_i ($i \in [k]$) to a distinct vertex in G. We consider two types of mappings.

Type 1: All Vertices of Q Mapped to Heavy Vertices. There are only $O((\sqrt{m})^k) = O(m^{k/2})$ such mappings because the number of heavy vertices is bounded by $O(\sqrt{m})$. For each mapping, we can check in constant time whether the subgraph of G induced by the mapping is indeed an occurrence of Q. The time of finding all such occurrences is therefore $O(m^{k/2})$.

Type 2: At least One Vertex of Q Mapped to a Light Vertex. W.l.o.g., assume that q_1 must be mapped to a light vertex. We consider mappings $f : V_Q \to V$ satisfying three conditions:

1. $f(q_1)$ is a light vertex;
2. $\{f(q_1), f(q_2)\}$ and $\{f(q_1), f(q_k)\}$ are edges in G;
3. for each $i \in \{2, 3, ..., (k-1)/2\}$, $\{f(q_{2i-1}), f(q_{2i})\}$ is an edge in G.

There are $O(m^{1.5})$ choices to satisfy conditions 1 and 2 (think: why?) and $O(m^{\frac{k-1}{2}-1})$ choices to satisfy condition 3. Hence, the total number of such mappings is bounded by $O(m^{k/2})$. For each mapping, we can check in constant time whether it induces an occurrence of Q. The time of finding all such occurrences is therefore $O(m^{k/2})$.

3 Enumerating Stars

Let Q be a k-star where k is a constant integer at least 1. It is rudimentary to find all its occurrences in a data graph G of m edges in $O(m^{\rho(Q)})$ time where $\rho(Q) = k$. The details are left to you.

4 Enumerating Arbitrary Subgraphs

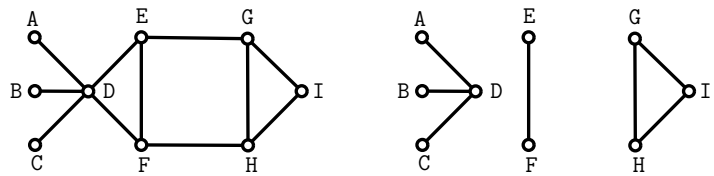
Let $Q = (V_Q, E_Q)$ be a pattern graph with a constant size. Given a fractional edge cover w of Q, we define

$$\textsc{supp}(w) \quad = \quad \{e \in E_Q \mid w(e) > 0\}. \tag{2}$$

For each vertex $u \in V_Q$, at least one edge in $\textsc{supp}(w)$ is incident on u (think: why?).

Theorem 3. *Every pattern graph Q admits an optimal fractional edge cover w^* whose $\textsc{supp}(w^*)$ induces a collection of vertex-disjoint odd-length cycles $\pentagon_1, ..., \pentagon_\alpha$ ($\alpha \geq 0$) and stars $\star_1, ..., \star_\beta$ ($\beta \geq 0$) such that $\rho(Q) = \sum_{i=1}^{\alpha} \rho(\pentagon_i) + \sum_{i=1}^{\beta} \rho(\star_i)$.*

We will prove the theorem in the next section. For an illustration, Figure 1(a) shows a pattern graph Q with $\rho(Q) = 5.5$, while Figure 1(b) shows the promised collection, which has $\alpha = 1$ odd-length cycle and $\beta = 2$ stars. It is easy to verify that the fractional edge cover numbers of the cycle and stars sum up to 5.5.



(a) The "fish" graph (b) The decomposition

Figure 1: Illustration of Theorem 3

Theorem 3 implies the following algorithm to find the occurrences of Q in $G = (V, E)$:

- For each $i \in [\alpha]$, find the set $M_i^{\pentagon}$ of occurrences of $\pentagon_i$.
- For each $i \in [\beta]$, find the set $M_i^{\star}$ of occurrences of $\star_i$.
- For each $(G_1^{\pentagon}, ..., G_\alpha^{\pentagon}, G_1^{\star}, ..., G_\beta^{\star}) \in M_1^{\pentagon} \times ... \times M_\alpha^{\pentagon} \times M_1^{\star} \times ... \times M_\beta^{\star}$, check in constant time whether the subgraphs $G_1^{\pentagon}, ..., G_\alpha^{\pentagon}, G_1^{\star}, ...,$ and $G_\beta^{\star}$ induce an occurrence of Q.

By the discussion in Sections 2 and 3, the running time is at the order of

$$\prod_{i=1}^{\alpha} m^{\rho(\pentagon_i)} \cdot \prod_{i=1}^{\beta} m^{\rho(\star_i)}$$

which is $O(m^{\rho(Q)})$ by Theorem 3.

As a side product, we have given a constructive proof of the AGM bound in Theorem 1. The algorithm is worst-case optimal because even just outputting the result takes $\Omega(m^{\rho(Q)})$ time by Theorem 2.

5 Proof of Theorem 3

We will prove the theorem through a series of lemmas about fractional edge covers. If w is a fractional edge cover of $Q = (V_Q, E_Q)$, we use $Q_{\text{SUPP}(w)}$ to denote the subgraph of Q induced by $\text{SUPP}(w)$.

Lemma 4. *Suppose that w^* is an optimal fractional edge cover of Q such that $Q_{\text{SUPP}(w^*)}$ is a collection of vertex-disjoint stars. Then, w^* maps each edge of $Q_{\text{SUPP}(w^*)}$ to 1 and every other edge of E_Q to 0.*

Proof. Trivial and omitted. □

Henceforth, a fractional edge cover w of Q is said to be *integral* if $w(e) \in \{0, 1\}$ for each $e \in E_Q$.

Lemma 5. *Suppose that $Q = (V_Q, E_Q)$ has an optimal fractional edge cover w^* such that $Q_{\text{SUPP}(w^*)}$ is a forest. Then, Q has an optimal fractional edge cover w such that*

- *w is integral;*
- *$Q_{\text{SUPP}(w)}$ induces a collection of vertex-disjoint stars.*

Proof. If $Q_{\text{SUPP}(w^*)}$ is a collection of vertex-disjoint stars, the claim follows immediately from Lemma 4. Next, we assume that this is not true.

Let $\mathcal{T}$ be an arbitrary tree in the forest $Q_{\text{SUPP}(w^*)}$ such that $\mathcal{T}$ is not a star. Root $\mathcal{T}$ at an arbitrary degree-1 node. Let z be a deepest leaf of $\mathcal{T}$ (i.e., its distance from the root is the largest among all leaves). Let u_1 be its parent. The value of $w^*(\{z, u_1\})$ must be 1. Let u_2 be the parent of u_1, and u_3 be the parent of u_2 (think: why can't u_1 and u_2 be the root?). We modify w^* as follows:

- increase $w^*(\{u_2, u_3\})$ by $w^*(\{u_1, u_2\})$;
- set $w^*(\{u_1, u_2\})$ to 0.

The new w^* is an optimal fractional edge cover of Q with one less edge in $\text{SUPP}(w^*)$ — because $\{u_1, u_2\}$ is no longer in $\text{SUPP}(w^*)$ — and $\text{SUPP}(w^*)$ still induces a forest.

If any tree in the forest $Q_{\text{SUPP}(w^*)}$ is still not a star, we can repeat the process to reduce $|\text{SUPP}(w^*)|$ further. Eventually, we obtain an optimal fractional edge cover w^* such that $Q_{\text{SUPP}(w^*)}$ is a collection of vertex-disjoint stars. □

Lemma 6. *Every bipartite $Q = (V_Q, E_Q)$ has an integral optimal fractional edge cover w^*.*

Proof. By Lemma 5, it suffices to prove that Q has an optimal fractional edge cover w^* whose $Q_{\text{SUPP}(w^*)}$ is a forest. If this is not true, then $Q_{\text{SUPP}(w^*)}$ must contain a cycle $\pentagon$. As Q is bipartite, $\pentagon$ must have an even number of edges, which we denote by $e_1, e_2, ..., e_k$ in some traversal order.

W.l.o.g., suppose that e_1 has the smallest weight under w^* among $e_1, e_2, ..., e_k$. Set $\Delta = w^*(e_1)$. We now modify w^* by

- decreasing the weights of $e_1, e_3, ..., e_{k-1}$ by Δ;
- increasing the weights of $e_2, e_4, ..., e_k$ by Δ.

The new w^* is an optimal fractional edge cover of Q whose $\text{SUPP}(w^*)$ has at least one less edge (because e_1 is no longer in $\text{SUPP}(w^*)$).

If the subgraph induced by $\text{SUPP}(w^*)$ still has a cycle, repeat the above process by further reducing $\text{SUPP}(w^*)$. This eventually gives a an optimal fractional edge cover w^* of Q whose $\text{SUPP}(w^*)$ induces a forest. □

Lemma 7. *Every pattern graph $Q = (V_Q, E_Q)$ has an optimal fractional edge cover w^* such that $w^*(e) \in \{0, 1/2, 1\}$ for each $e \in E_Q$.*

Proof. Build a graph $Q' = (V_{Q'}, E_{Q'})$ as follows. Each vertex $u \in V_Q$ defines two vertices u_L and u_R to $V_{Q'}$ — refer to them as the *left copy* and *right copy* of u, respectively. Each edge $\{u, v\} \in E_Q$ defines two edges $\{u_L, v_R\}$ and $\{v_L, u_R\}$ to $E_{Q'}$. As every edge of $E_{Q'}$ connects a left-copy vertex to a right-copy vertex, Q' is bipartite. By Lemma 6, Q' has an integral optimal fractional edge cover w^*.

Define a function $w : E_Q \to \mathbb{R}_{\geq 0}$ where

$$w(\{u, v\}) = \frac{1}{2}(w^*(\{u_L, v_R\}) + w^*(\{v_L, u_R\}))$$

for each $\{u, v\} \in E_Q$. Note that $w(\{u, v\}) \in \{0, 1/2, 1\}$.

We claim that w is a fractional edge cover of Q. Indeed, for any vertex $u \in V_Q$, it holds that

$$\begin{aligned}
& \sum_{v \in V_Q : \{u,v\} \in E_Q} w(\{u, v\}) \\
= \; & \frac{1}{2} \sum_{v \in V_Q : \{u,v\} \in E_Q} w^*(\{u_L, v_R\}) + w^*(\{v_L, u_R\}) \\
= \; & \frac{1}{2} \sum_{v_R \in V_{Q'} : \{u_L, v_R\} \in E_{Q'}} w^*(\{u_L, v_R\}) + \frac{1}{2} \sum_{v_L \in V_{Q'} : \{v_L, u_R\} \in E_{Q'}} w^*(\{v_L, u_R\}) \\
\geq \; & 1
\end{aligned}$$

where the last step used the fact that w^* is a fractional edge cover of Q'.

The total weight of w equals half of the total weight of w^*. We argue that w must be an optimal fractional edge cover of Q. Otherwise, let w_0 be an optimal fractional edge cover of Q with a lower total weight. Construct a function $w_0^* : E_{Q'} \to \mathbb{R}_{\geq 0}$ by setting, for each $\{u, v\} \in E_Q$:

$$w_0^*(\{u_L, v_R\}) = w_0^*(\{u_R, v_L\}) = w_0(\{u, v\}).$$

It is easy to check that w_0^* is a fractional edge cover of Q' whose total weight is twice the total weight of w_0 and, hence, is less than the total weight of w^*, giving a contradiction. □

Henceforth, a fractional edge cover w of Q is said to be *half-integral* if $w(e) \in \{0, 1/2, 1\}$ for each $e \in E_Q$.

Lemma 8. *Any pattern graph $Q = (V_Q, E_Q)$ has an optimal fractional edge cover w^* such that*

- *w^* is half-integral;*
- *$Q_{\text{SUPP}(w^*)}$ is a collection of vertex-disjoint odd-length cycles and stars.*

Proof. Let w^* be a half-integral optimal fractional edge cover promised by Lemma 7. Suppose that $Q_{\text{SUPP}(w^*)}$ contains an even-length cycle ⬠. Let us denote its edges by $e_1, e_2, ..., e_k$ in some traversal order. The weight $w^*(e_i)$ must be $1/2$ for all $i \in [k]$ (think: why?). We now modify w^* by

- decreasing the weights of $e_1, e_3, ..., e_{k-1}$ by $1/2$;
- increasing the weights of $e_2, e_4, ..., e_k$ by $1/2$.

This creates another half-integral optimal fractional edge cover but with fewer edges in $\text{SUPP}(w^*)$. By repeating the process, we obtain a half-integral optimal fractional edge cover w^* such that $Q_{\text{SUPP}(w^*)}$ has no even-length cycles.

We call an edge $\{u, v\} \in Q_{\text{SUPP}(w^*)}$ a *pendant edge* if u appears in a cycle ⬠ of $G_{\text{SUPP}(w^*)}$ where v does not belong. Let us traverse ⬠ from u, and denote by $e_1, e_2, ..., e_k$ the edges encountered (in this order); in other words, e_1 and e_k are incident on u. Note that k is an odd number (because $Q_{\text{SUPP}(w^*)}$ contains no more even-length cycles). The weight $w^*(e_i)$ must be $1/2$ for all $i \in [k]$ (think: why?).

We argue that the weight of $\{u, v\}$ under w^* must be $1/2$. Otherwise, its weight must be 1 (because w^* is half-integral). In that case, we can construct another fractional edge cover of Q with a *smaller* total weight, which contradicts the optimality of w^*. For this purpose, we modify w^* as follows to reduce its total weight:

- decrease $w^*(e)$ by 1/2 for $e \in \{e_1, e_3, ..., e_k\}$ — after which, $w^*(e) = 0$;
- increase $w^*(e)$ by 1/2 for $e \in \{e_2, e_4, ..., e_{k-1}\}$ — after which, $w^*(e) = 1$.

The resulting w^* is still a fractional edge cover of Q, but its total weight has decreased by 1/2.

Now that we know $w^*(\{u, v\}) = 1/2$, we proceed to modify w^* as follows:

- increase $w^*(\{u, v\})$ by 1/2;
- decrease $w^*(e)$ by 1/2 for $e \in \{e_1, e_3, ..., e_k\}$ — after which, $w^*(e) = 0$;
- increase $w^*(e)$ by 1/2 for $e \in \{e_2, e_4, ..., e_{k-1}\}$ — after which, $w^*(e) = 1$.

After modification, w^* remains as a half-integral optimal edge cover of Q but has fewer edges in $\textsc{supp}(w^*)$. By repeating the process, we obtain a half-integral optimal fractional edge cover w^* such that $Q_{\textsc{supp}(w^*)}$ has no even-length cycles and no pendent edges.

At this time, $Q_{\textsc{supp}(w^*)}$ is a collection of odd-length cycles $\pentagon_1, ..., \pentagon_\alpha$ ($\alpha \geq 0$) and a forest $\mathcal{T}$ such that $\pentagon_1, ..., \pentagon_\alpha$ and $\mathcal{T}$ are vertex disjoint. Using the procedure in the proof of Lemma 5, we can further modify w^* by adjusting only the weights on the edges of $\mathcal{T}$ so that

- w^* as a half-integral optimal fractional edge cover;
- $Q_{\textsc{supp}(w^*)}$ is a collection of $\pentagon_1, ..., \pentagon_\alpha$ and stars $\star_1, ..., \star_\beta$ such that
  - $\pentagon_1, ..., \pentagon_\alpha$, $\star_1, ..., \star_\beta$ are vertex disjoint;
  - the union of the vertices in $\star_1, ..., \star_\beta$ is the vertex set of $\mathcal{T}$.

This w^* satisfies the requirements of Lemma 8. □

Theorem 3 is a corollary of Lemma 8.

6 Remarks

Theorem 1 is a special case of a result presented in [2]. The proof of Theorem 3 was adapted from an argument in [1].

References

[1] Sepehr Assadi, Michael Kapralov, and Sanjeev Khanna. A simple sublinear-time algorithm for counting arbitrary subgraphs via edge sampling. In *Innovations in Theoretical Computer Science (ITCS)*, pages 6:1–6:20, 2019.

[2] Albert Atserias, Martin Grohe, and Daniel Marx. Size bounds and query plans for relational joins. *SIAM Journal on Computing*, 42(4):1737–1767, 2013.